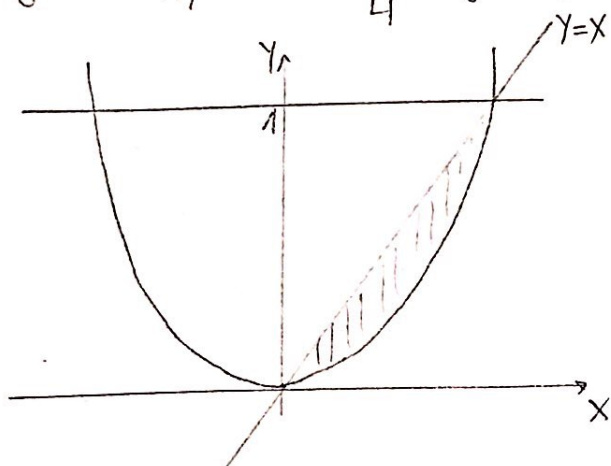


Ej $F_{xy}(x,y) = \frac{21}{4} x^2 y \mathbb{1}_{\{x^2 < y \leq 1\}}$ * hallar $P(X > Y)$



$$P(X > Y) = \int_{-1}^1 \int_{x^2}^1 \frac{21}{4} x^2 y \, dy \, dx$$

2.1) $\Omega = \{1, 2, 3, 4, 5, 6\}$ $\mathcal{A} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$

$X: \Omega \rightarrow \mathbb{R}$

a) $X(\omega) = \omega$ NO ES VARIABLE ALEATORIA

b) $X(\omega) = \mathbb{1}_{\{\omega \text{ es par}\}}$ ES VARIABLE ALEATORIA

c) $X(\omega) = \mathbb{1}_{\{\omega \in \{1, 4\}\}}$ NO ES VARIABLE ALEATORIA

Una VA. es un $F: \Omega \rightarrow \mathbb{R}$ / las ocurrencias de los valores de la función corresponden a eventos en $\mathcal{A}$ y, por lo tanto, se les puede asignar probabilidad

2.2) a)

$$b) P(-2 < X \leq 2) = F_X(2) - F_X(-2) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(-2 \leq X \leq 2) = F_X(2) - F_X(-2) + P(X = -2) = 1 - \frac{1}{3} + \frac{1}{3} = 1$$

$$P(-2 \leq X < 2) = F_X(2) - F_X(-2) + P(X = -2) - P(X = 2) \\ = 1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{3} = \frac{2}{3}$$

$$P(-2 < X < 2) = F_X(2) - F_X(-2) - P(X = 2) = 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$$

$$c) P(X \in (-2, -1)) = P(-2 < X < -1) = 0$$

$$P(|X| \leq 1) = P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{6} dx = \frac{1}{3}$$

$$P(X \in (1, 2)) = P(1 < X < 2) = 0$$

$$d) P(X \leq 1.5 | X < 2) = \frac{P(X \leq 1.5 \wedge X < 2)}{P(X < 2)} = \frac{P(X \leq 1.5)}{P(X < 2)} =$$

$$= \frac{F_X(1.5)}{F_X(2) - P(X = 2)} = \frac{2/3}{1 - 1/3} = 1$$

$$P(X \leq 1,5 | X \leq 2) = \frac{P(X \leq 1,5)}{P(X \leq 2)} = \frac{F_X(1,5)}{F_X(2)} = \frac{2/3}{1} = \frac{2}{3}$$

$$e) P(X = -2 | |X| = 2) = \frac{P(X = -2)}{P(|X| = 2)} = \frac{1/3}{2/3} = \frac{1}{2}$$

2.3) 3 verdes
5 rojas

X : "cant. de bolas verdes observables"

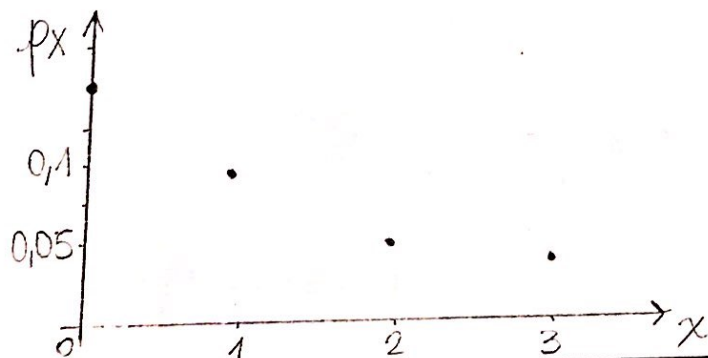
$$a) p_X(0) = P(X=0) = \frac{5^4}{8^4} \approx 0,15$$

$$p_X(1) = P(X=1) = \frac{5^3 \cdot 3}{8^4} \approx 0,092$$

$$p_X(2) = P(X=2) = \frac{5^2 \cdot 3^2}{8^4} \approx 0,055$$

$$p_X(3) = P(X=3) = \frac{5 \cdot 3^3}{8^4} \approx 0,033$$

$$p_X(4) = 0$$



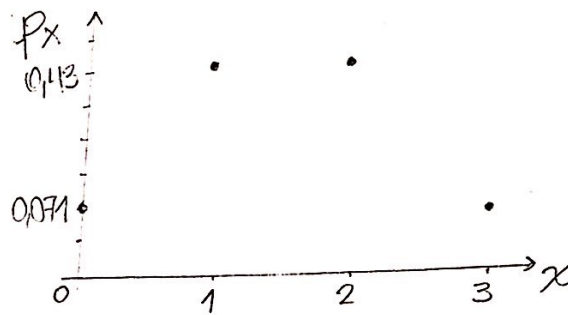
$$b) p_X(0) = \frac{\binom{5}{4}}{\binom{8}{4}} \approx 0,071$$

$$p_X(1) = \frac{\binom{5}{3}\binom{3}{1}}{\binom{8}{4}} \approx 0,43$$

$$p_X(2) = \frac{\binom{5}{2}\binom{3}{2}}{\binom{8}{4}} \approx 0,43 \quad p_X(3) = \frac{\binom{5}{1}\binom{3}{3}}{\binom{8}{4}} \approx 0,071$$

d) $p_{(N)}$

$$p_X(4) = 0$$



2.4) $p = \frac{5}{8}$ de salir "cara"

a) N_k : "cant de lanzamientos hasta k-ésima cara"

$$p_N(k) = \binom{n-1}{k-1} \left(\frac{5}{8}\right)^k \left(\frac{3}{8}\right)^{n-k} \quad (\text{V.A. DE PASCAL})$$

$$b) P(N_1 \text{ es par}) = \left(\frac{5}{8}\right) \sum_{n=0}^{\infty} \left(\frac{3}{8}\right)^{2n+1} = \frac{15}{64} \left(\frac{1}{1 - 9/64}\right) = \frac{3}{11}$$

$$c) P(N_1=3) = \left(\frac{3}{8}\right)^2 \cdot \frac{5}{8} \approx 0,088$$

$$P(N_2=5 | N_1=2) = P(N_1=3) \approx 0,088$$

$$d) P(N_1 > 3) = \left(\frac{3}{8}\right)^3 \approx 0,053$$

$$P(N_1 > 5 | N_1 > 2) = P(N_1 > 3) \approx 0,053$$

$$e) P(N_2 > 3) = \binom{3}{1} \left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right) \approx 0,26$$

$$P(N_2 > 5 | N_2 > 2) = P(N_2 > 3) \approx 0,26$$

2.5) N : cant de particulas alfa emitidas (p/segundo)
 $N \sim \text{POISSON}(1/2)$

$$a) P(N > 3) = 1 - P(N \leq 3) = 1 - [P(N=0) + P(N=1) + P(N=2) + P(N=3)]$$

$$= 1 - \left[\frac{(1/2)^0 e^{-1/2}}{0!} + \frac{(1/2)^1 e^{-1/2}}{1!} + \frac{(1/2)^2 e^{-1/2}}{2!} + \frac{(1/2)^3 e^{-1/2}}{3!} \right] = 0,00175$$

$$b) P(N \text{ impar}) = \sum_{n=0}^{\infty} \frac{(1/2)^{2n+1} e^{-1/2}}{(2n+1)!} = \frac{e^{-1/2}}{2} \sum_{n=0}^{\infty} \frac{(1/4)^n}{(2n+1)!} \approx 0,316$$

$$\sum_{n=0}^{\infty} \frac{(1/2)^{2k+1} e^{-1/2}}{(2k+1)!} = \frac{\sin(1/2)}{e^{1/2}}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

↑
Fórmula de Euler

2.6) debo contar en $(0, 1/4)$ o $(1/4, 1)$

X : "longitud de corte" $\sim U(0, 1)$

$$P(0 < X < 1/4 \cup 3/4 < X < 1) = \frac{1/4 + 1/4}{1} = \frac{1}{2}$$

2.7) X VA. con $F_X(x) = 2x \mathbb{1}_{\{0 \leq x \leq 1\}}$

$$P(X_1 = 2 | X_1 + X_2 = 3) = \frac{P(X_1 = 2 \wedge X_1 + X_2 = 3)}{P(X_1 + X_2 = 3)}$$

$$= \frac{\int_{0.21}^{0.22} 2x \, dx}{\int_{0.08}^{0.09} 2x \, dx + \int_{0.12}^{0.13} 2x \, dx + \int_{0.21}^{0.22} 2x \, dx + \int_{0.30}^{0.31} 2x \, dx} \approx 0.32$$

$$\int_{0.08}^{0.09} 2x \, dx + \int_{0.12}^{0.13} 2x \, dx + \int_{0.21}^{0.22} 2x \, dx + \int_{0.30}^{0.31} 2x \, dx$$

2.8) T_k : "tiempo (en seg) en emitir k partículas alfa"

$$f_{T_k}(t) = \frac{(1/2)^k}{(k-1)!} t^{k-1} e^{-t/2} \mathbb{1}_{\{t \geq 0\}} \quad T_k \sim \text{GAMMA}(k, 1/2)$$

$$a) P(T_1 > 3) = 1 - P(T_1 \leq 3) = 1 - \int_0^3 \frac{(1/2)}{0!} t^0 e^{-t/2} dt =$$

$$= 1 - \int_0^3 \frac{1}{2} e^{-t/2} dt \approx 0.22$$

$$P(T_1 > 5 | T_1 > 2) \stackrel{??}{=} \text{ver carpeta} P(T_1 > 3) \approx 0.22$$

$$b) P(T_3 > 3) = 1 - P(T_3 \leq 3) = 1 - \int_0^3 \frac{(1/2)^3}{2!} t^2 e^{-t/2} dt = \\ \cong 0,81$$

$$P(T_3 > 5 | T_3 > 2) = P(T_3 > 3) \cong 0,81$$

$$2.9) \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$Z \sim N(0,1)$ $\leftarrow$ normal estandar

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$a) \begin{array}{ll} \Phi(1) = 0,8413 & \Phi(2) = 0,9772 \\ \Phi(0) = 0,5 & \Phi(3) = 0,9987 \\ \Phi(4) = 1 & \Phi(5) = \Phi(6) = 1 \end{array}$$

$$b) \Phi^{-1}(0,5) = 0 \quad \Phi^{-1}(0,75) = 0,6745$$

$$\Phi^{-1}(0,9) = 1,2816 \quad \Phi^{-1}(0,95) = 1,6449$$

$$\Phi^{-1}(0,99) = 2,3263 \quad \Phi^{-1}(0,995) = 2,5758$$

$$\Phi^{-1}(0,999) = 3,0902$$

$$c) P(-0,43 < Z < 1,32) = 0,4066 - (1 - 0,6664) = 0,573$$

$$P(1,28 < Z < 1,64) = 0,9495 - 0,8997 = 0,0498$$

$$P(|Z| < 1,64) = 0,9495 - (1 - 0,9495) = 0,899$$

$$d) P(Z < a) = 0,05 \Rightarrow a = -1,6449$$

$$P(Z > b) = 0,1 \Rightarrow b = -1,2816$$

$$P(|Z| < c) = 0,95 \Rightarrow c = 1,96$$

$$2.10) Z \sim N(0,1), X = Z^2, f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$a) \Phi(3) = P(Z \leq 3)$$

$$F_X(x) = P(X \leq x) = P(Z^2 \leq x)$$

$$= P(|Z| \leq \sqrt{x})$$

$$= P(-\sqrt{x} \leq Z \leq \sqrt{x})$$

$$= \Phi(\sqrt{x}) - \Phi(-\sqrt{x}) = F_X(x)$$

$$= 2\Phi(\sqrt{x}) - 1 = F_X(x)$$

$$\int_{-\infty}^{\sqrt{x}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt + \int_{-\infty}^{-\sqrt{x}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

ESTA BIEN
definida

$$\int_{-\infty}^{\sqrt{x}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = 1 - \int_{\sqrt{x}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

$$\int_{-\infty}^{-\sqrt{x}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \int_{\sqrt{x}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

regla de la cadena

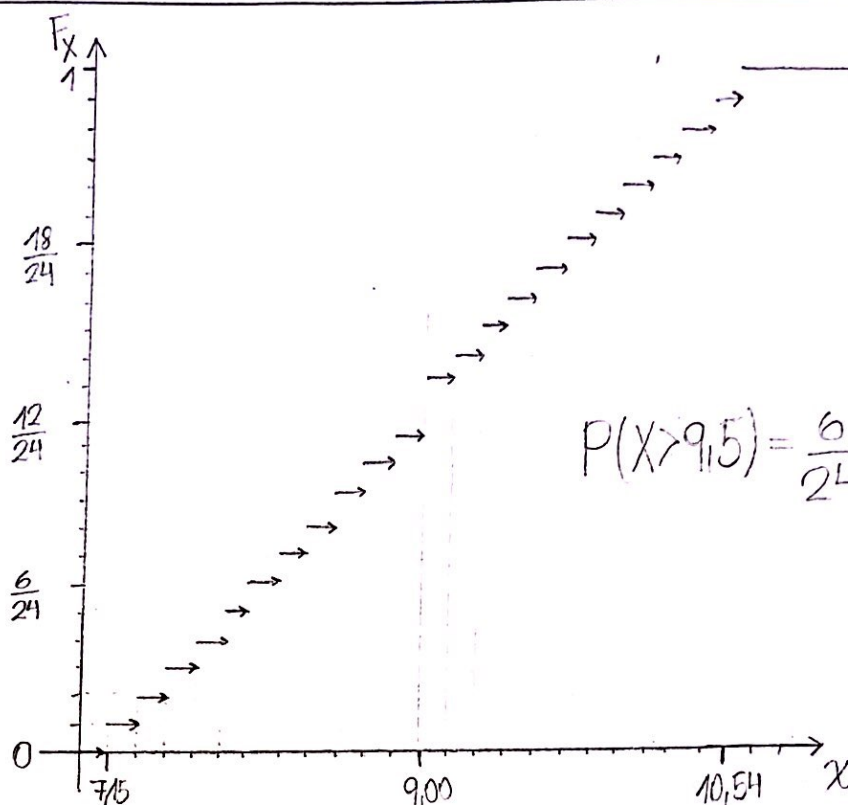
$$b) f_X(x) = F'_X(x) = 2\Phi'(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{2\pi x}} e^{-x/2}$$

$$c) \int_0^{\infty} x^{-1/2} e^{-x} dx = \int_0^{\infty} \frac{1}{\sqrt{2}} y^{-1/2} e^{-y/2} \frac{1}{\sqrt{2}} dy = \frac{1}{\sqrt{2}} \sqrt{2\pi} = \sqrt{\pi}$$

$y = 2x$
 $dy = 2dx$
 $x^{-1/2} = \sqrt{2} y^{-1/2}$

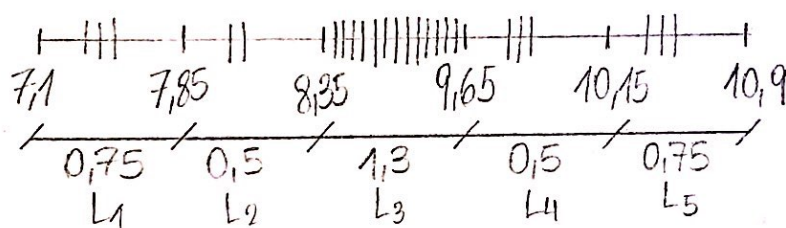
2.11)

a)



$$P(X > 9,5) = \frac{6}{24} = 0,25$$

b)



$$p_1 = \frac{3}{24} \quad p_2 = \frac{2}{24} \quad p_3 = \frac{13}{24} \quad p_4 = \frac{3}{24} \quad p_5 = \frac{3}{24}$$

$$h(x) = \frac{1}{6} \mathbb{1}\{7,1 < x \leq 7,85\} + \frac{1}{6} \mathbb{1}\{7,85 < x \leq 8,35\} + \frac{5}{12} \mathbb{1}\{8,35 < x \leq 9,65\} + \frac{1}{4} \mathbb{1}\{9,65 < x \leq 10,15\} + \frac{1}{6} \mathbb{1}\{10,15 < x \leq 10,9\}$$

$$P(X > 9,5) = \frac{5}{12} (9,65 - 9,5) + \frac{1}{4} (10,15 - 9,65) + \frac{1}{4} (10,9 - 10,15) = 0,375$$

2.12) T = duración en años del tiempo de trabajo sin fallos

$$F_T(t) = 1 - e^{-t} \mathbb{1}_{\{t \geq 0\}}$$

$$a) T^* = \min(T, 1) \begin{cases} T \leq 1 \\ 1 \leq T \end{cases}$$

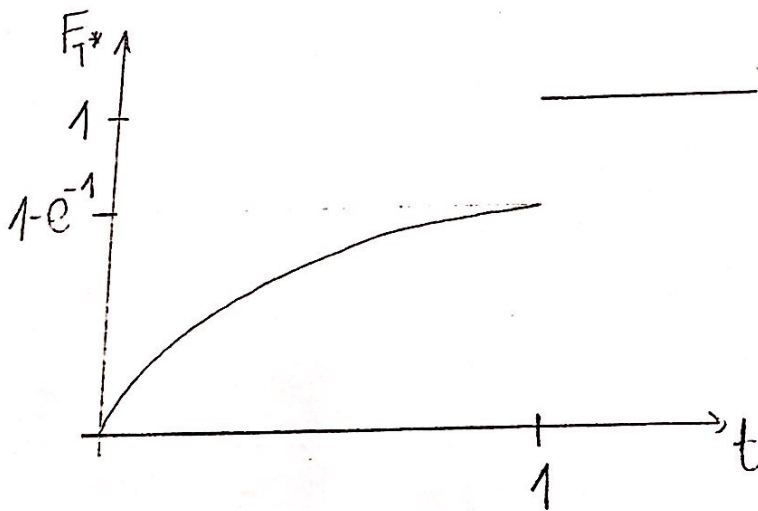
$$P(T^* \leq t) = F_{T^*}(t)$$

$$I) t < 0 \rightarrow P(T^* \leq t) = 0$$

$$II) t \geq 1 \rightarrow P(T^* \leq t) = 1$$

$$III) 0 \leq t < 1 \rightarrow P(T^* \leq t) = P(T \leq t) = 1 - e^{-t}$$

$$F_{T^*}(t) = \begin{cases} 0 & \mathbb{1}_{\{t < 0\}} \\ 1 - e^{-t} & \mathbb{1}_{\{0 \leq t < 1\}} \\ 1 & \mathbb{1}_{\{t \geq 1\}} \end{cases} \Rightarrow F_{T^*}(t) = e^{-t} \mathbb{1}_{\{0 \leq t < 1\}}$$



$$b) P(T^* = 1) = 1 - (1 - e^{-1}) = e^{-1} \approx 0,368$$

2.13) X.V.A. / $F_X(x) = P(X \leq x)$

a) $T = G(X)$

$$F_T(t) = P(T \leq t) = P(G(X) \leq t) = P(X \leq G^{-1}(t)) = F_X(G^{-1}(t))$$

b) $U = F_X(X) \Rightarrow U \sim U(0,1) \Rightarrow F_U(u) = \begin{cases} u & \{0 < u < 1\} \\ 1 & \{u \geq 1\} \end{cases}$

c) $V = G^{-1}(F_X(X))$

$$\begin{aligned} F_V(v) &= P(V \leq v) = P(G^{-1}(F_X(X)) \leq v) = P(F_X(X) \leq G(v)) = \\ &= P(X \leq F_X^{-1}(G(v))) = F_X(F_X^{-1}(G(v))) = G(v) \end{aligned}$$

2.14) $F_X(x) = 1 - e^{-2x}$

$F_X(X) \sim U(0,1)$

$X_1 = \frac{7}{4} \Rightarrow F_X\left(\frac{7}{4}\right) = 1 - e^{-2(7/4)} = 0,9698$

$X_2 = \frac{111}{60} \Rightarrow F_X\left(\frac{111}{60}\right) = 1 - e^{-2(111/60)} = 0,975$

$X_3 = \frac{137}{60} \Rightarrow F_X\left(\frac{137}{60}\right) = 1 - e^{-2(137/60)} = 0,9896$

2.15) $U \sim U(0,1)$ $X = h(U)$

hallar $h / F_{h(U)}(u) = F_X(x)$

$$F_X(x) = \frac{1}{3} \mathbb{1}_{\{-2 \leq x < -1\}} + \left(\frac{1}{6}x + \frac{1}{2}\right) \mathbb{1}_{\{-1 \leq x < 1\}} + \frac{2}{3} \mathbb{1}_{\{1 \leq x < 2\}}$$

$$F_{h(U)}(x) = F_X(x) = F_U(h^{-1}(x))$$

$$+ \mathbb{1}_{\{x \geq 2\}}$$

$$F_U(F_X^{-1}(x)) = F_X^{-1}(x)$$

$$F_U(u) = u \mathbb{1}_{\{0 \leq u < 1\}} + \mathbb{1}_{\{u \geq 1\}}$$

$$X = F_X^{-1}(F_U(u))$$

$$F_U(u) = F_X(x)$$

versión alternativa $X \sim U$
/ en la carpeta practice

$$X = F_X^{-1}(F_U(u))$$

2.16) $T^* = h(x)$

$$X \sim \exp(\lambda = 1/2)$$

$$F_T^*(T^*) = F_X(x)$$

$$F_X(x) = 1 - e^{-\lambda x} \mathbb{1}_{\{x \geq 0\}}, \lambda = \frac{1}{2}$$

$$T^* = F_{T^*}^{-1}(F_X(x))$$

$$F_{T^*}^{-1}(u) = \begin{cases} -\ln(1-u) & 0 < u < 1 - e^{-1} \\ 1 & 1 - e^{-1} \leq u < 1 \end{cases} \quad \lambda = \frac{1}{2}$$

$$\begin{aligned} u &= 1 - e^{-t} \\ e^{-t} &= 1 - u \\ t &= -\ln(1-u) \end{aligned}$$

$$h(x) = \begin{cases} -\ln(e^{-\lambda x}) & \text{si } 0 < 1 - e^{-\lambda x} < 1 - e^{-1} \\ 1 & \text{si } 1 - e^{-1} \leq 1 - e^{-\lambda x} < 1 \end{cases} \quad \Leftrightarrow \begin{cases} x & 0 < x < 2 \\ 1 & 2 \leq x \end{cases}$$

$$\begin{aligned} 1 &> e^{-\lambda x} > e^{-1} \\ 0 &> -\frac{x}{2} > -1 \end{aligned}$$

$$\begin{aligned} 1 - e^{-1} &\leq 1 - e^{-\lambda x} < 1 \\ e^{-1} &\geq e^{-\lambda x} > 0 \end{aligned}$$

$$0 < \frac{x}{2} < 1$$

$$1 \geq -\lambda x > -\infty$$

$$0 < x < 2$$

$$2 \leq x$$

2.17)

$$a) F_T(t) = 1 - e^{-\int_0^t \frac{\beta}{\alpha} \left(\frac{u}{\alpha}\right)^{\beta-1} du} \quad \forall \{t > 0\}$$

$$F_T(t) = 1 - e^{-\frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta}} = 1 - e^{-\left(\frac{t}{\alpha}\right)^{\beta}} \quad \forall \{t > 0\}$$

$$F_T(t) = F'_T(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^{\beta}} \quad \forall \{t > 0\}$$

b) T: "tiempo hasta pinchadura de neumático por objeto punzante"

⇒

T: "tiempo hasta que se desgaste el surco y pierde agarre"

⇒

T: "tiempo hasta que revienta el neumático por falla de fábrica"

⇒

T: "tiempo hasta que el usuario se da cuenta que salió rallada"

⇒

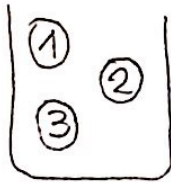
T: "tiempo hasta que falla el sistema de enfriamiento"

⇒

T: "tiempo hasta que el motor se quema por un cambio de tensión"

⇒

5



se extraen 2 sin reposición

X: "número de primer bolilla"

Y: "número de segunda bolilla"

a) prob. conjunta y marginales

b) $P(X < Y)$

c) ¿son independientes?

a)

$Y \backslash X$	1	2	3	p_Y
1	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{6}$
2	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{2}{6}$
3	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{2}{6}$
p_X	$\frac{2}{6}$	$\frac{2}{6}$	$\frac{2}{6}$	1

$$b) P(X < Y) = \frac{1}{6}$$

$$c) p_{XY}(2,1) = \frac{1}{6}$$

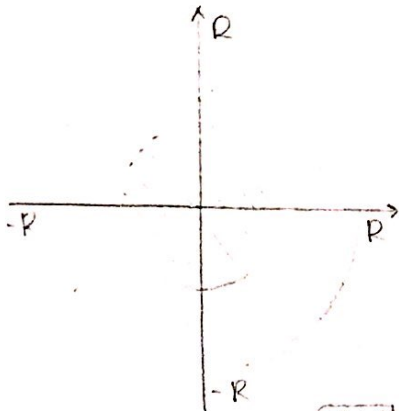
$$p_X(2) = \frac{2}{6}$$

$$p_Y(1) = \frac{2}{6}$$

$$p_{XY}(2,1) \neq p_X(2) \cdot p_Y(1)$$

$\Rightarrow$ no son independientes

Ej



DIST. UNIFORME

$$F_{XY}(x,y) = \frac{1}{|A|} \mathbb{1}\{x^2 + y^2 \leq R^2\}$$

$$P(X^2 + Y^2 < (\frac{R}{2})^2) = P((X,Y) \in A) = \frac{|A|}{|A|}$$

$$F_X(x) = \int_{-\infty}^{+\infty} F_{XY} dy = \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \frac{1}{|A|} dy = \frac{2\sqrt{R^2-x^2}}{\pi R^2} \mathbb{1}\{-R < x < R\}$$

$$F_Y(y) = \int_{-\infty}^{+\infty} F_{XY} dx = \int_{-\sqrt{R^2-y^2}}^{\sqrt{R^2-y^2}} \frac{1}{|A|} dx = \frac{2\sqrt{R^2-y^2}}{\pi R^2} \mathbb{1}\{-R < y < R\}$$

si el soporte es rectangular no es necesariamente independiente
pero si no es rectangular no es independiente

$\Rightarrow$ X e Y no son independientes en este caso

Ej $T \sim \text{EXP}(1)$ $F_T(t) = \lambda e^{-\lambda t} \mathbb{1}\{t > 0\} = e^{-t} \mathbb{1}\{t > 0\}$

simular 5 valores de $T^* = \min(T, 1)$

$$F_T(t) = F_{T^*}(t^*) \quad (\text{mirar 2.12})$$

$$U \sim U(0,1)$$

$$u_1 = 0,433 = 1 - e^{-t_1^*} \rightarrow t_1^* = \bigcirc$$

$$u_2 = 0,933 = 1 - e^{-t_2^*} \rightarrow t_2^* = 1$$

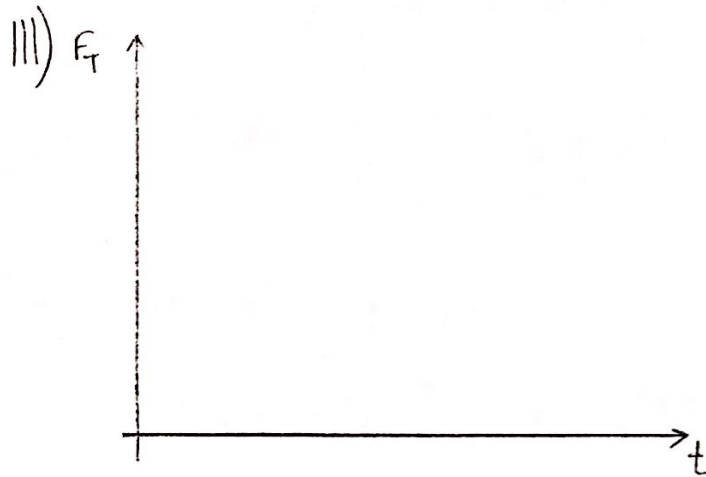
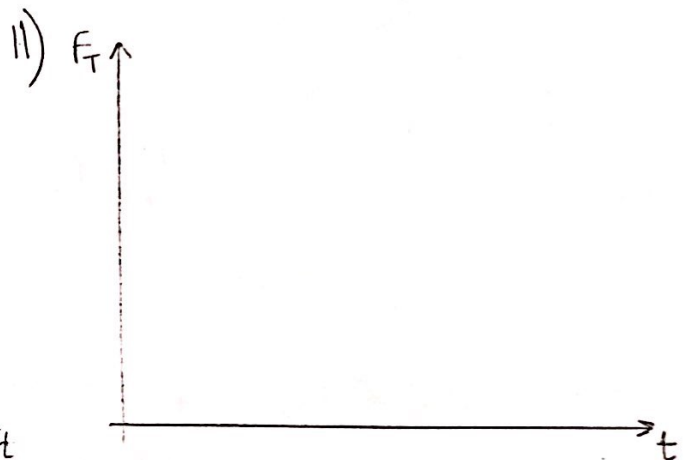
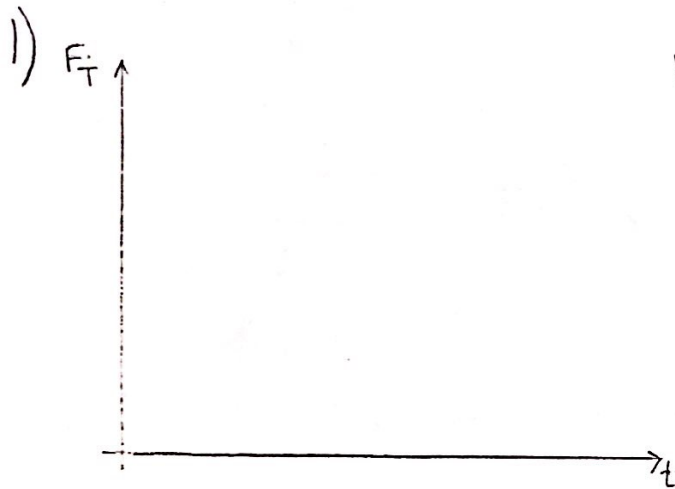
idem con u_3, u_4, u_5

Wzrostka (2.17)

$$c) i) \alpha = 1, \beta = 0,5 \Rightarrow F_T(t) = -\frac{1}{2} t^{-1/2} e^{t/2} \mathbb{1}_{\{t > 0\}}$$

$$ii) \alpha = 1, \beta = 1 \Rightarrow F_T(t) = -e^t \mathbb{1}_{\{t > 0\}}$$

$$iii) \alpha = 1, \beta = 1,5 \Rightarrow F_T(t) = -\frac{3}{2} t^{1/2} e^{t/2} \mathbb{1}_{\{t > 0\}}$$



$$d) P(T > 1) = 1 - P(T \leq 1) =$$

$$i) P(T > 4 | T > 3) = \frac{P(T > 4)}{P(T > 3)} =$$

$$II) P(T > 1) = 1 - P(T \leq 1) =$$

$$P(T > 4 | T > 3) = \frac{P(T > 4)}{P(T > 3)} =$$

$$III) P(T > 1) = 1 - P(T \leq 1) =$$

$$P(T > 4 | T > 3) = \frac{P(T > 4)}{P(T > 3)} =$$

$$2.18) F_T(t) = (1 - e^{-\sqrt{t/60}}) \mathbb{1}_{\{t > 0\}}$$

$$P(T < 60 | T > 30) = \frac{P(T < 60, T > 30)}{P(T > 30)} = \frac{P(30 < T < 60)}{P(T > 30)} =$$

$$= \frac{F_T(60) - F_T(30)}{1 - F_T(30)} \approx 0,254$$

2.14) X : número de alumnos

$$F_X(x) = \frac{2x}{225} \mathbb{1}_{\{0 < x < 15\}}$$

$$\begin{aligned} a) F_{X|X \in (3,12)}(x) &= \frac{F_X(x) \mathbb{1}_{\{x \in (3,12)\}}}{P(X \in (3,12))} = \\ &= \frac{\frac{2x}{225} \mathbb{1}_{\{x \in (3,12)\}}}{\int_3^{12} \frac{2x}{225} dx = \frac{x^2}{225} \Big|_3^{12}} = \\ &= \frac{\frac{2x}{225} \mathbb{1}_{\{x \in (3,12)\}}}{\frac{3}{5}} = \frac{2x}{135} \mathbb{1}_{\{x \in (3,12)\}} \end{aligned}$$

$$\begin{aligned} b) F_{X|X \notin (3,12)}(x) &= \frac{F_X(x) \mathbb{1}_{\{x \notin (3,12)\}}}{P(X \notin (3,12))} = \\ &= \frac{\frac{2x}{225} \mathbb{1}_{\{x \notin (3,12)\}}}{\int_0^3 \frac{2x}{225} dx + \int_{12}^{15} \frac{2x}{225} dx} = \\ &= \frac{x}{115} \mathbb{1}_{\{x \notin (3,12)\}} \end{aligned}$$

2.20) X : "distancia del pto de impacto al centro de un blanco circular"

$$F_X(x) = \frac{2}{7} \mathbb{1}\{0 \leq x < 2\} + \frac{10-2x}{21} \mathbb{1}\{2 \leq x < 5\}$$

$$a) F_{X|X \in (0,3)}(x) = \frac{F_X(x) \mathbb{1}\{X \in (0,3)\}}{P(X \in (0,3))} =$$

$$= \frac{\frac{2}{7} \mathbb{1}\{0 \leq x < 2\} + \frac{10-2x}{21} \mathbb{1}\{2 \leq x < 3\}}{\int_0^2 \frac{2}{7} dx + \int_2^3 \frac{10-2x}{21} dx} =$$

$$= \frac{6}{17} \mathbb{1}\{0 \leq x < 2\} + \left(\frac{10}{17} - \frac{2x}{17}\right) \mathbb{1}\{2 \leq x < 3\}$$

$$b) F_{X|X \in (3,5)}(x) = \frac{F_X(x) \mathbb{1}\{X \in (3,5)\}}{P(X \in (3,5))} =$$

$$= \frac{\frac{10-2x}{21} \mathbb{1}\{3 < x < 5\}}{\int_3^5 \frac{10-2x}{21} dx} =$$

$$= \left(\frac{5}{2} - \frac{x}{2}\right) \mathbb{1}\{3 < x < 5\}$$

2.21)

3 VERDES
2 AMARILLAS
3 ROJAS

→ 4 bolas

$P(X,Y)$

a) X : "cant. de bolas verdes observables"

Y : "cant. de bolas amarillas observables"

SIN reposición

$Y \backslash X$	0	1	2	3	4	p_Y
0	0	$\frac{3}{70}$	$\frac{9}{70}$	$\frac{3}{70}$	0	$\frac{3}{14}$
1	$\frac{1}{35}$	$\frac{9}{35}$	$\frac{9}{35}$	$\frac{1}{35}$	0	$\frac{4}{7}$
2	$\frac{3}{70}$	$\frac{9}{70}$	$\frac{3}{70}$	0	0	$\frac{3}{14}$
3	0	0	0	0	0	0
4	0	0	0	0	0	0
p_X	$\frac{1}{14}$	$\frac{3}{7}$	$\frac{3}{7}$	$\frac{1}{14}$	0	1

$$P_{XY}(1,0) = \frac{\binom{3}{1} \binom{2}{0} \binom{3}{3}}{\binom{8}{4}} = \frac{3}{70}$$

$$P_{XY}(2,0) = \frac{\binom{3}{2} \binom{2}{0} \binom{3}{2}}{\binom{8}{4}} = \frac{9}{70}$$

$$P_{XY}(3,0) = \frac{\binom{3}{3} \binom{2}{0} \binom{3}{1}}{\binom{8}{4}} = \frac{3}{70}$$

$$P_{XY}(0,1) = \frac{\binom{3}{0} \binom{2}{1} \binom{3}{3}}{\binom{8}{4}} = \frac{1}{35}$$

$$P_{XY}(1,1) = \frac{\binom{3}{1} \binom{2}{1} \binom{3}{2}}{\binom{8}{4}} = \frac{9}{35}$$

$$P_{XY}(2,1) = \frac{\binom{3}{2} \binom{2}{1} \binom{3}{1}}{\binom{8}{4}} = \frac{9}{35}$$

$$P_{XY}(3,1) = \frac{\binom{3}{3} \binom{2}{1} \binom{3}{0}}{\binom{8}{4}} = \frac{1}{35}$$

$$P_{XY}(0,2) = \frac{\binom{3}{0} \binom{2}{2} \binom{3}{2}}{\binom{8}{4}} = \frac{3}{70}$$

$$P_{XY}(1,2) = \frac{\binom{3}{1} \binom{2}{2} \binom{3}{1}}{\binom{8}{4}} = \frac{9}{70}$$

$$P_{XY}(2,2) = \frac{\binom{3}{2} \binom{2}{2} \binom{3}{0}}{\binom{8}{4}} = \frac{3}{70}$$

$$P(X+Y \leq 2) = \frac{1}{2}$$

b) CON REPOSICIÓN

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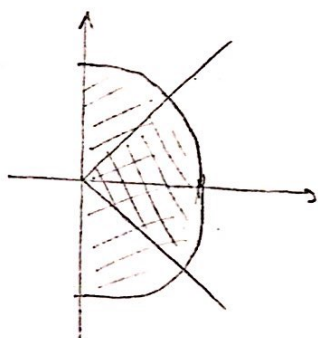
X \ Y	0	1	2	3	4	$p_{y\cdot}$
0	$\frac{81}{4096}$	$\frac{81}{4096}$	$\frac{81}{4096}$	$\frac{81}{4096}$	$\frac{81}{4096}$	$\frac{405}{4096}$
1	$\frac{27}{2048}$	$\frac{27}{2048}$	$\frac{27}{2048}$	$\frac{27}{2048}$	0	$\frac{27}{512}$
2	$\frac{9}{1024}$	$\frac{9}{1024}$	$\frac{9}{1024}$	0	0	$\frac{27}{1024}$
3	$\frac{3}{512}$	$\frac{3}{512}$	0	0	0	$\frac{3}{256}$
4	$\frac{1}{256}$	0	0	0	0	$\frac{1}{256}$
$p_{x\cdot}$	$\frac{21}{4096}$	$\frac{195}{4096}$	$\frac{171}{4096}$	$\frac{135}{4096}$	$\frac{81}{4096}$	$\frac{793}{4096}$

$$P(X+Y \leq 2) = \frac{387}{4096}$$

2.22) $(X, Y) \sim U(\Lambda)$, $\Lambda = \{(x, y) \in \mathbb{R}^2 / x^2 + y^2 \leq 4, x \geq 0\}$

a) $P(|Y| < X) = \frac{\left(\frac{\pi 2^2}{4}\right)}{\left(\frac{\pi 2^2}{2}\right)} = \frac{1}{2}$

$$F_{xy}(x, y) = \frac{1}{|\Lambda|} = \frac{1}{2\pi}$$



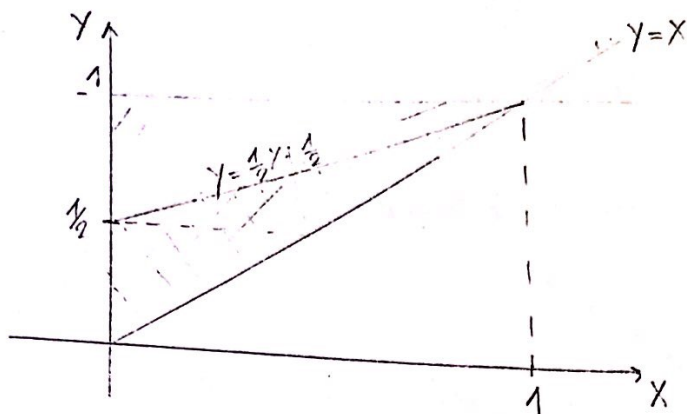
b) $F_X(x) = \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \frac{1}{2\pi} dy = \frac{2\sqrt{4-x^2}}{2\pi} \mathbb{1}_{\{0 \leq x < 2\}}$

$$F_Y(y) = \int_0^{\sqrt{4-y^2}} \frac{1}{2\pi} dx = \frac{\sqrt{4-y^2}}{2\pi} \mathbb{1}_{\{-2 \leq y \leq 2\}}$$

c) $F_{xy}(x, y) \neq F_X(x) \cdot F_Y(y) \Rightarrow$ no son independientes

2.23) $F_{XY}(x,y) = 8xy \mathbb{1}_{\{0 \leq x \leq y \leq 1\}}$

a) $P(X+1 > 2Y)$



$$\begin{aligned}
 P(X+1 > 2Y) &= \int_0^1 \int_x^{1/2x+1/2} 8xy \, dy \, dx = \int_0^1 4x \left(\frac{1}{2}x + \frac{1}{2} - x \right) dx = \\
 &= \int_0^1 4x \left(-\frac{1}{2}x + \frac{1}{2} \right) dx = \int_0^1 (-2x^2 + 2x) dx = \\
 &= \left[-\frac{2}{3}x^3 + x^2 \right]_0^1 = -\frac{2}{3} + 1 = \frac{1}{3} \approx 0.33
 \end{aligned}$$

b) $F_X(x) = \int_x^1 F_{XY}(x,y) \, dy \mathbb{1}_{\{0 < y < 1\}} = 4x - 4x^3$

$F_Y(y) = \int_0^y F_{XY}(x,y) \, dx \mathbb{1}_{\{0 < y < 1\}} = 4y^3$

c) NO SON INDEP. PORQUE EL SOPORTE NO ES RECTANGULAR

2.24) $f_{xy}(x,y) = \frac{1}{\pi\sqrt{3}} e^{-\frac{2}{3}(x^2 - xy + y^2)}, \{x,y\} \in \mathbb{R}^2$

a) $x^2 - xy + y^2 = (y - \frac{1}{2}x)^2 - \frac{1}{4}x^2 + x^2 = (y - \frac{1}{2}x)^2 + \frac{3}{4}x^2$

$\Rightarrow -\frac{2}{3}(y - \frac{1}{2}x)^2 - \frac{2}{3} \cdot \frac{3}{4}x^2 = -\frac{2}{3}(y - \frac{1}{2}x)^2 - \frac{x^2}{2}$

$\Rightarrow \frac{1}{\pi\sqrt{3}} e^{-\frac{2}{3}(y - \frac{1}{2}x)^2 - \frac{x^2}{2}} = \frac{1}{\pi\sqrt{3}} e^{-\frac{2}{3}(y - \frac{1}{2}x)^2} \cdot e^{-\frac{x^2}{2}}$

quiero algo de la forma

$\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} \Rightarrow \frac{1}{\pi\sqrt{3}} e^{-\frac{(y - \frac{1}{2}x)^2}{3/2}} \cdot e^{-\frac{x^2}{2}}$

$= \frac{1}{\pi\sqrt{3}} e^{-\frac{1}{2} \frac{(y - \frac{1}{2}x)^2}{3/4}} \cdot e^{-\frac{x^2}{2}} \rightarrow \sigma = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

$= \frac{1}{2\pi\sqrt{\frac{3}{4}}} e^{-\frac{1}{2} \frac{(y - \frac{1}{2}x)^2}{3/4}} \cdot e^{-\frac{x^2}{2}} = \frac{\sqrt{2\pi}}{2\pi} \underbrace{\frac{1}{\sqrt{2\pi} \cdot \frac{\sqrt{3}}{2}} e^{-\frac{1}{2} \frac{(y - \frac{1}{2}x)^2}{3/4}} \cdot e^{-\frac{x^2}{2}}}_{N \sim (\frac{1}{2}x, \frac{3}{4})}$

$\Rightarrow f_x(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}} \Rightarrow f_y(y) = \frac{e^{-y^2/2}}{\sqrt{2\pi}}$

b) $\Rightarrow$ NO SON INDEP PORQUE $f_{xy}(x,y) \neq f_x(x) \cdot f_y(y)$

2.25) X : "cant. de fallas de tejido" $\sim \text{Poisson}(2)$

Y : "cant. de fallas de tejido" $\sim \text{Poisson}(4)$

X e Y i.i.d.

a) calcular la prob. de que un rollo no tenga fallas

$$P_X(x) = P(X=x) = \frac{2^x e^{-2}}{x!} \quad P_Y(y) = P(Y=y) = \frac{4^y e^{-4}}{y!}$$

$$P(X=0, Y=0) = P(X=0) \cdot P(Y=0)$$

$$P(X=0) = \frac{2^0 e^{-2}}{0!} = e^{-2} \quad P(Y=0) = \frac{4^0 e^{-4}}{0!} = e^{-4}$$

$$P(X=0, Y=0) = e^{-2} \cdot e^{-4} = 0,00248$$

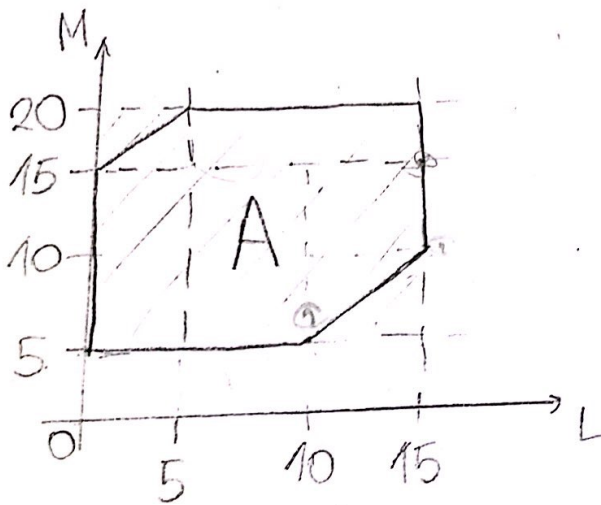
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$$b) P(X=1 \cup Y=1) = \frac{2^1 e^{-2}}{1!} + \frac{4^1 e^{-4}}{1!} \approx 0,37$$

$$c) P(X=1 | X=1 \cup Y=1) = \frac{P(X=1, Y=1)}{P(X=1 \cup Y=1)} = \frac{2e^{-2} \cdot 4e^{-4}}{2e^{-2} + 4e^{-4}} \approx 0,058$$

2.26) L : "horario de llegada de Lucas" $\sim U(0, 15) + 15$
 M : "horario de llegada de Monk" $\sim U(5, 20) + 5$

$$P(\text{"se encuentran"}) = P((L, M) \in A) = \frac{\frac{25}{2} + 50 + 100 + 25 + \frac{25}{2}}{15 \times 15}$$



$$= \frac{8}{9} \cong 0,89$$